

▼ PDS Practical-2

Name: Jafar Razzak Birbal

Div:SY-A

Roll No:A-72

Aim: To perform data ingestion and cleaning using Pandas.

Theory: Data ingestion is the process of collecting and importing data from various sources into a system for analysis. These sources can include databases, CSV files, APIs, web scraping, or real-time streaming data. The main goal of data ingestion is to gather raw data and store it in a structured format so it can be easily accessed and analyzed. In Python, libraries like Pandas are commonly used to load data using functions such as `read_csv()`, `read_excel()`, or `read_json()`.

Double-click (or enter) to edit

```
import pandas as pd
df = pd.read_csv("https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv")
df.head()
df.to_csv("titanic.csv", index=False)
df = pd.read_csv("titanic.csv")
df.head()
df.info()
df.head()
df.tail()
df.isnull().sum()
# Fill missing Age with average
df["Age"] = df["Age"].fillna(df["Age"].mean())
# Drop Cabin column (too many missing)
df = df.drop("Cabin", axis=1)
df = df.drop_duplicates()
df.isnull().sum()
df.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  -
 0   PassengerId   891 non-null    int64
 1   Survived      891 non-null    int64
 2   Pclass        891 non-null    int64
 3   Name          891 non-null    object
 4   Sex           891 non-null    object
 5   Age           714 non-null    float64
 6   SibSp         891 non-null    int64
 7   Parch         891 non-null    int64
 8   Ticket        891 non-null    object
 9   Fare          891 non-null    float64
10   Cabin         204 non-null    object
11   Embarked      889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily	female	35.0	1	0	113803	53.1000	S

Q 1). What is data ingestion and why is it important?

-> Answer: Data ingestion is the process of collecting and importing data from different sources such as CSV files, databases, or APIs into a system for analysis.

Importance:

- It is the first step in data science
- Makes data available for processing
- Ensures data is organized and accessible
 - Library Used: The most commonly used Python library for data ingestion and cleaning is Pandas.

Q 2. Why is data cleaning necessary? -> Answer: Data cleaning is necessary to improve the quality of data before analysis or machine learning.

- Reasons:

Removes missing values

Fixes incorrect or inconsistent data

Removes duplicate entries

Ensures accurate results

- Without cleaning, analysis can give wrong results.

Q 3. How do you load a CSV file into a Pandas DataFrame?

-> Answer: A CSV file can be loaded into a Pandas DataFrame using the `read_csv()` function. *Example: `import pandas as pd`
`df = pd.read_csv("file_name.csv")`

- This reads the CSV file and stores it in a DataFrame.

Q4. Difference between `head()` and `tail()`

The `head()` function in Pandas displays the first five rows of a DataFrame by default, which is useful for a quick initial inspection of the data. On the other hand, the `tail()` function displays the last five rows of a DataFrame by default, which is useful for checking the end of the dataset, especially after operations that might append data.

Both functions are excellent for quickly getting a sense of your data's structure and content without printing the entire DataFrame.

Q 5. How to identify missing values?

-> Answer: Missing values in a dataset can be identified using the `isnull()` function.

- Example: `df.isnull().sum()`
- Explanation:
 - `isnull()` → returns True/False for missing values
 - `sum()` → counts total missing values in each column

**** Conclusion ****

Conclusion

This practical successfully demonstrated the process of data ingestion and cleaning using the Pandas library in Python. We covered essential steps including:

- **Data Loading:** Ingesting data from a CSV file into a Pandas DataFrame.
- **Initial Data Inspection:** Using `head()`, `tail()`, and `info()` to understand the dataset's structure and contents.
- **Missing Value Identification:** Utilizing `isnull().sum()` to pinpoint missing data across columns.
- **Data Cleaning Techniques:** Applying methods to handle missing values by filling them (e.g., with the mean) and dropping columns with excessive missing data (`Cabin` column).
- **Duplicate Removal:** Ensuring data uniqueness by dropping duplicate rows.

Through these steps, the raw dataset was transformed into a cleaner, more usable format, ready for further analysis or machine learning tasks. This practical highlights the importance of thorough data preparation as a foundational step in any data science workflow.